Comprehensive Analysis of DNA Methylation

Using the Illumina HumanMethylation450 BeadChip Platform

Group 7

DNA/RNA Dynamics Course

Università di Bologna

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1 Introduction

DNA methylation is a central epigenetic modification in which a methyl group is covalently added to the 5-carbon position of cytosine, predominantly within CpG dinucleotides. It contributes to the regulation of gene expression, genomic imprinting, X-chromosome inactivation and the silencing of repetitive and transposable elements, and its dysregulation is a well-established feature of many pathological states, including cancer.

This project analyses DNA methylation data generated with the Illumina HumanMethylation450 BeadChip (the “450k” array), which interrogates more than 450,000 CpG sites distributed across promoters, gene bodies, CpG islands, shores and shelves throughout the human genome [1]. The array combines two distinct probe chemistries, Infinium Type I and Type II, whose differing dynamic ranges must be accounted for during normalisation [2].

For each CpG site the platform yields two complementary measures of methylation. The beta value is the ratio of methylated signal to total signal and ranges from 0 (fully unmethylated) to 1 (fully methylated), providing a biologically interpretable scale; the M value is the log2 ratio of methylated to unmethylated intensities and has more favourable statistical properties (approximate homoscedasticity) for differential analysis [3]. Both metrics are used at different stages of this pipeline.

The aim of the study is to compare a control group (CTRL) and a disease group (DIS) and to identify differentially methylated positions (DMPs) between them. The complete workflow, raw-data import, quality control, normalisation, principal component analysis, statistical testing and visualisation, was implemented in R using the *minfi* Bioconductor package [4], following the methodology of the DNA/RNA Dynamics course at the University of Bologna.

The dataset comprises 8 samples (4 CTRL and 4 DIS) profiled on the 450k array; the sample sheet is given in Table 1. Several pipeline steps use parameters assigned specifically to Group 7: the manifest address 54740449 (Step 3), a detection p-value threshold of 0.05 (Step 5), *preprocessQuantile* normalisation (Step 7) and the t-test for differential methylation (Step 9).

Table 1. Sample sheet for Group 7.

SampleID	Sex	Group	Sentrix_ID	Sentrix_Position
GSM5319592	Male	CTRL	200121140049	R01C02
GSM5319603	Male	DIS	3999356129	R02C02
GSM5319604	Female	DIS	3999547012	R02C01
GSM5319607	Male	CTRL	3999547012	R03C02
GSM5319609	Female	DIS	3999547016	R02C01
GSM5319613	Female	DIS	3999547017	R02C01
GSM5319615	Female	CTRL	3999547017	R04C01
GSM5319616	Female	CTRL	3999547017	R06C01

2 Analytical Pipeline and R Code

STEP 0. Basic Preparation (Environment Setting and Importing Libraries)

The working directory is set to the folder containing the raw data, and the required libraries are loaded. (Only packages actually used in the pipeline are loaded.)

```
rm(list=ls())
setwd("~/your_working_directory")
suppressMessages(library(minfi))
library(gplots)
library(qqman)
library(VennDiagram)
library(sva)
```

STEP 1. Data Import and Object Creation

Raw IDAT data are loaded with *minfi* to build an RGChannelSet object (RGset). The sample sheet is read to obtain the file paths, and read.metharray.exp() then loads all IDAT files.

```
list.files("Input_Data/")

SampleSheet <- read.csv("Input_Data/SampleSheet_Report_II.csv", header=T)
SampleSheet

baseDir <- ("Input_Data")
targets <- read.metharray.sheet(baseDir)
targets

RGset <- read.metharray.exp(targets=targets)
RGset
save(RGset, file="RGset.RData")
```

STEP 2. Red and Green Channels Extraction

The Red and Green fluorescence-intensity matrices are extracted from the RGChannelSet. Rows correspond to array addresses and columns to samples.

```
Red <- data.frame(getRed(RGset))
dim(Red)
head(Red)

Green <- data.frame(getGreen(RGset))
dim(Green)
head(Green)
```

STEP 3. Probe Information and Address Check (Group 7: address 54740449)

The Red and Green fluorescence values at the assigned address 54740449 are extracted across all 8 samples, and the Illumina 450k manifest is queried to determine the probe type and colour channel .

```
# Fluorescence values at address 54740449
Red[rownames(Red)=="54740449",]
Green[rownames(Green)=="54740449",]

# Check probe type in the manifest
load("Illumina450Manifest_clean.RData")
Illumina450Manifest_clean[Illumina450Manifest_clean$AddressA_ID=="54740449",]
Illumina450Manifest_clean[Illumina450Manifest_clean$AddressB_ID=="54740449",]
```

The manifest lookup reveals whether address 54740449 is the AddressA (present in both Type I and Type II probes) or AddressB (present only in Type I probes). Type I probes use two addresses (A and B) for unmethylated and methylated signals respectively, measured in a single channel (Red or Green depending on Color_Channel). Type II probes use a single address and emit in both channels simultaneously. The results obtained for Group 7 are reported in the table below.

Table 2. Red and Green fluorescence values at address 54740449 for all 8 samples.

Sample	Red	Green	Type	Color Channel
GSM5319592_200121140049_R01C02	160	127	I	Red
GSM5319603_3999356129_R02C02	255	67	I	Red
GSM5319604_3999547012_R02C01	408	130	I	Red
GSM5319607_3999547012_R03C02	599	217	I	Red
GSM5319609_3999547016_R02C01	181	61	I	Red
GSM5319613_3999547017_R02C01	305	72	I	Red
GSM5319615_3999547017_R04C01	537	133	I	Red
GSM5319616_3999547017_R06C01	423	98	I	Red

Address 54740449 is the AddressB of probe cg21802055, confirming it is a Type I probe measured in the Red channel (Color_Channel = Red). For Type I probes, the AddressA carries the unmethylated signal and the AddressB carries the methylated signal, both measured in the same channel. The Green channel values at this address therefore represent out-of-band signals.

STEP 4. Creation of the MSet.raw Object

The preprocessRaw() function converts the RGChannelSet into a MethylSet object (MSet.raw), in which each probe's methylated and unmethylated signals are computed by combining the Red and Green channel intensities according to the Infinium design type of each probe.

```
MSet.raw <- preprocessRaw(RGset)
MSet.raw
save(MSet.raw, file="MSet_raw.RData")

Meth <- as.matrix(getMeth(MSet.raw))
head(Meth)
Unmeth <- as.matrix(getUnmeth(MSet.raw))
head(Unmeth)
```

STEP 5. Quality Control

a. QC Plot

The getQC() function computes the median log2 methylated (mMed) and unmethylated (uMed) signal intensities for each sample, and plotQC() displays them on a scatter plot. Samples above the threshold line (mMed > 10.5 and uMed > 10.5) are considered to be of good quality.

```
qc <- getQC(MSet.raw)
qc
plotQC(qc)
```

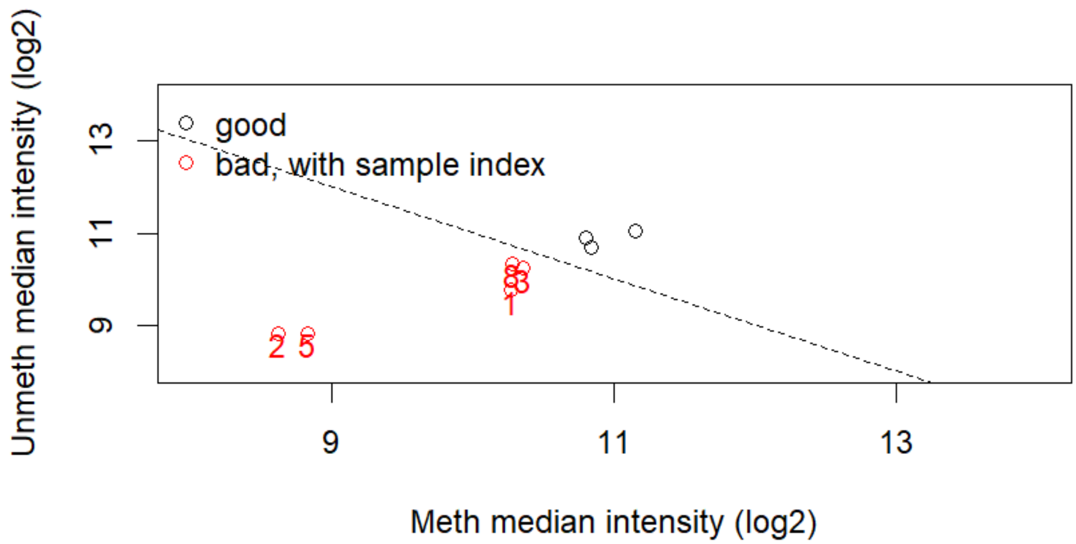

Figure 1. QC plot showing median methylated vs. unmethylated signal intensity per sample. Samples above the diagonal threshold line are of good quality

Console output (getQC):

```
DataFrame with 8 rows and 2 columns
                          mMed       uMed
200121140049_R01C02     10.263      9.763
3999356129_R02C02        8.622      8.820
3999547012_R02C01       10.351     10.237
3999547012_R03C02       10.794     10.903
3999547016_R02C01        8.827      8.833
3999547017_R02C01       10.837     10.695
3999547017_R04C01       11.150     11.046
3999547017_R06C01       10.278     10.333
```

Comment. plotQC() flags a sample as bad when the mean of its median methylated (mMed) and median unmethylated (uMed) log2 intensities is below the default cutoff of 10.5. On this criterion five samples are flagged (red, with index in Figure 1): sample 1 (mean 10.01), sample 2 (8.72), sample 3 (10.29), sample 5 (8.83) and sample 8 (10.31); only samples 4, 6 and 7 lie above the line. The cutoff is conservative for this dataset, and all 8 samples were retained because the more informative per-sample detection p-value profiles (Step 5c) show failure fractions of only 0.04–0.56%, far below the 5% level normally used to exclude a sample.

b. Negative Control Probe Intensities

Negative control probes measure non-specific background fluorescence. Their signal intensities, measured in both the Red and Green channels, are expected to remain below log2(1000) ≈ 10. The controlStripPlot() function visualises these intensities for each sample.

```
df_TypeControl <- data.frame(getProbeInfo(RGset, type="Control"))
str(df_TypeControl)
table(df_TypeControl$Type)

controlStripPlot(RGset, controls="NEGATIVE")
```

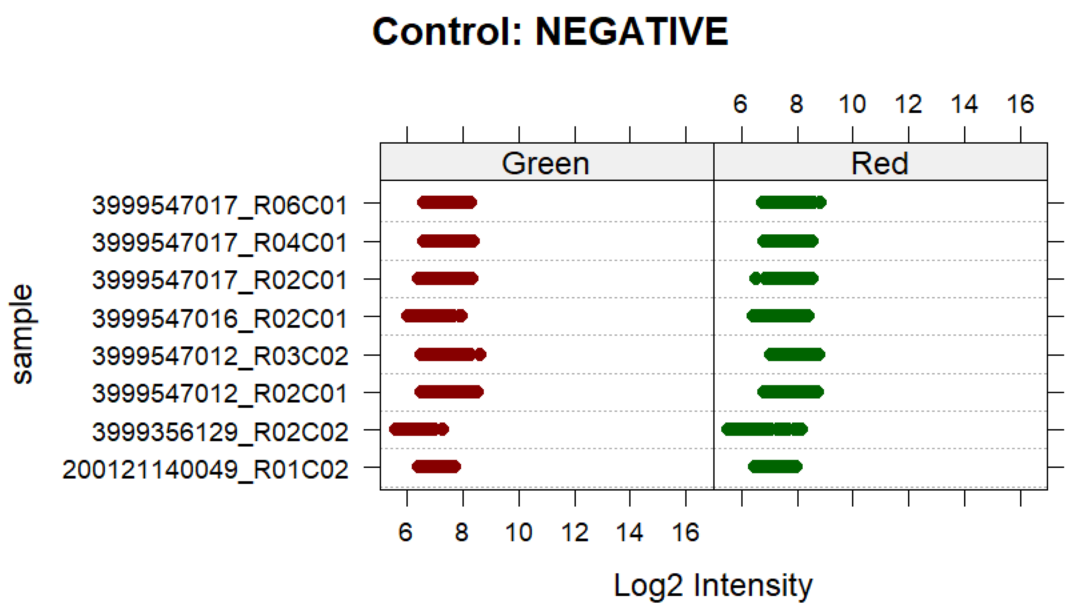

Figure 2. Control strip plot for NEGATIVE control probes. All signals should be below log2 intensity = 10 in both channels

Comment. The negative-control intensities lie below log2 ≈ 10 in both the Red and Green channels for every sample (Figure 2), the expected behaviour for background signal. Note that in R ≥ 4.0.0 there is a known minor minfi bug by which the Red and Green axis labels are swapped in controlStripPlot().

c. Detection P-values (Group 7 threshold: 0.05)

The detectionP() function computes a detection p-value for each probe in each sample by comparing the probe's signal against the distribution of negative control probe signals. A probe is considered to have failed quality control if its detection p-value exceeds the assigned threshold. Group 7 uses a threshold of 0.05.

```
detP <- detectionP(RGset)
save(detP, file="detP.RData")

dim(detP)
head(detP)

# Logical matrix: TRUE = failed probe (detP > 0.05)
failed <- detP > 0.05
table(failed)
summary(failed)
```

```
# Fraction of failed probes per sample
means_of_columns <- colMeans(failed)
means_of_columns
```

Table 3. Number of failed positions per sample (detection p-value > 0.05).

Sample	Failed positions (detP > 0.05)	Fraction
GSM5319592_200121140049_R01C02	202	0.04%
GSM5319603_3999356129_R02C02	306	0.06%
GSM5319604_3999547012_R02C01	1156	0.24%
GSM5319607_3999547012_R03C02	366	0.08%
GSM5319609_3999547016_R02C01	2735	0.56%
GSM5319613_3999547017_R02C01	460	0.09%
GSM5319615_3999547017_R04C01	384	0.08%
GSM5319616_3999547017_R06C01	674	0.14%

Comment. Across the 8 samples there are 6,283 probe positions with a detection p-value above the Group 7 threshold of 0.05. The number of failed positions per sample ranges from 202 (GSM5319592) to 2,735 (GSM5319609), i.e. failure fractions of 0.04–0.56% (colMeans of the failed matrix). As all values are well below the 5% threshold commonly used for exclusion, all 8 samples are retained for downstream analysis.

STEP 6. Calculation of Beta and M Values, Density Plots

Raw beta and M values are extracted from MSet.raw using getBeta() and getM() respectively. Samples are then divided into CTRL and DIS groups according to the sample sheet, and the mean methylation (beta) and mean M values are calculated per probe across samples within each group. Kernel density estimates are plotted to compare the two groups.

```
beta <- getBeta(MSet.raw)
M    <- getM(MSet.raw)

pheno <- read.csv("Input_Data/SampleSheet_Report_II.csv",
                  header=T, stringsAsFactors=T)
pheno$Group

# Subset by group
beta_CTRL <- beta[, pheno$Group=="CTRL"]
beta_DIS  <- beta[, pheno$Group=="DIS"]
M_CTRL    <- M[,    pheno$Group=="CTRL"]
M_DIS     <- M[,    pheno$Group=="DIS"]

# Mean per probe across samples
mean_of_beta_CTRL <- apply(beta_CTRL, 1, mean, na.rm=T)
mean_of_beta_DIS  <- apply(beta_DIS,  1, mean, na.rm=T)
mean_of_M_CTRL    <- apply(M_CTRL,    1, mean, na.rm=T)
mean_of_M_DIS     <- apply(M_DIS,     1, mean, na.rm=T)

# Density distributions
d_mean_of_beta_CTRL <- density(mean_of_beta_CTRL, na.rm=T)
d_mean_of_beta_DIS  <- density(mean_of_beta_DIS,  na.rm=T)
d_mean_of_M_CTRL    <- density(mean_of_M_CTRL,    na.rm=T)
d_mean_of_M_DIS     <- density(mean_of_M_DIS,     na.rm=T)

par(mfrow=c(1,2))
```

```r
plot(d_mean_of_beta_CTRL, main="Density of Beta Values", col="blue")
lines(d_mean_of_beta_DIS, col="red")
legend("topright", legend=c("CTRL","DIS"), col=c("blue","red"), lty=1)

plot(d_mean_of_M_CTRL, main="Density of M Values", col="blue")
lines(d_mean_of_M_DIS, col="red")
legend("topright", legend=c("CTRL","DIS"), col=c("blue","red"), lty=1)
par(mfrow=c(1,1))
```

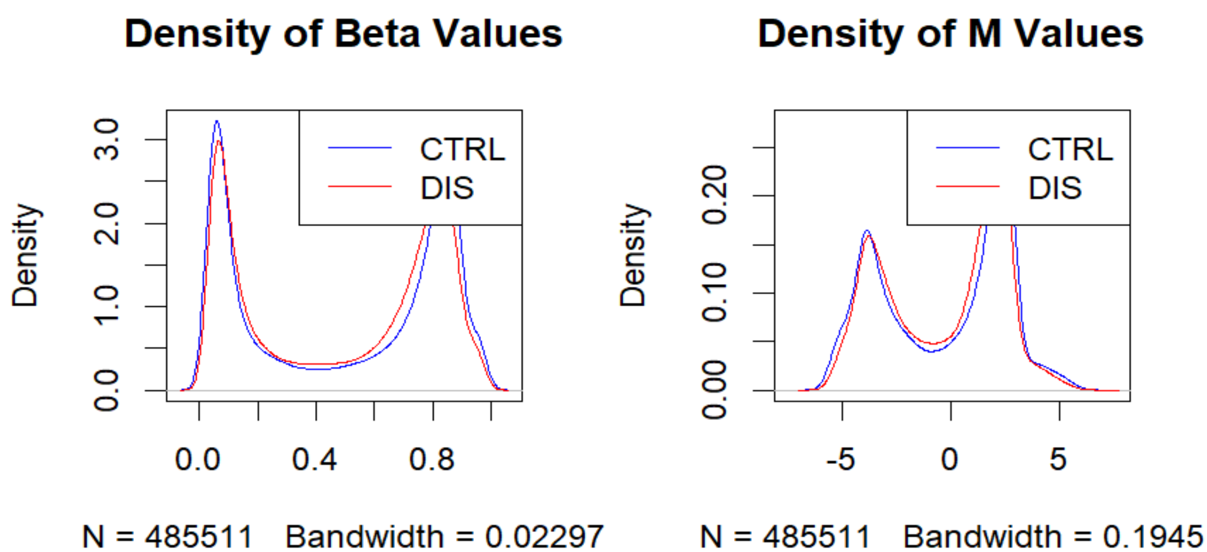

Figure 3. Density plots of mean raw beta values (left) and mean M values (right) for CTRL (blue) and DIS (red) groups

Comment. The mean-beta distributions of the two groups are almost superimposable (Figure 3) and both show the characteristic bimodal 450k shape, with an unmethylated peak near $\beta \approx 0.1$ and a methylated peak near $\beta \approx 0.9$. The M-value densities (Figure 4) are correspondingly bimodal around $M \approx -3.5$ and $M \approx +2.5$. The absence of a global shift between CTRL and DIS indicates that any methylation differences are likely confined to specific loci rather than affecting the genome as a whole, and will require per-probe testing to detect. Beta values are reported for interpretability, while M values are preferred where homoscedasticity matters [3].

STEP 7. Data Normalization with preprocessQuantile

Group 7 applies preprocessQuantile normalization. This method performs stratified quantile normalization separately for Type I and Type II probes, correcting for differences in signal distribution between the two probe chemistries. The effect of normalization is assessed via a 6-panel comparison plot.

```r
# Load manifest and subset by probe type
load("Illumina450Manifest_clean.RData")
dfI  <-
Illumina450Manifest_clean[Illumina450Manifest_clean$Infinium_Design_Type=="I",]
dfI  <- droplevels(dfI)
dfII <-
Illumina450Manifest_clean[Illumina450Manifest_clean$Infinium_Design_Type=="II",
]
dfII <- droplevels(dfII)
```

```
# Subset raw beta by probe type
beta_I  <- beta[rownames(beta) %in% dfI$IlmnID,]
beta_II <- beta[rownames(beta) %in% dfII$IlmnID,]

mean_of_beta_I  <- apply(beta_I,  1, mean, na.rm=T)
mean_of_beta_II <- apply(beta_II, 1, mean, na.rm=T)
sd_of_beta_I    <- apply(beta_I,  1, sd,   na.rm=T)
sd_of_beta_II   <- apply(beta_II, 1, sd,   na.rm=T)

d_mean_of_beta_I  <- density(mean_of_beta_I,  na.rm=T)
d_mean_of_beta_II <- density(mean_of_beta_II, na.rm=T)
d_sd_of_beta_I    <- density(sd_of_beta_I,    na.rm=T)
d_sd_of_beta_II   <- density(sd_of_beta_II,   na.rm=T)

# Apply preprocessQuantile
preprocessQuantile_results <- preprocessQuantile(RGset)
beta_preprocessQuantile <- getBeta(preprocessQuantile_results)
save(beta_preprocessQuantile, file="beta_preprocessQuantile.RData")

# Subset normalized beta by probe type
beta_preprocessQuantile_I  <-
beta_preprocessQuantile[rownames(beta_preprocessQuantile) %in% dfI$IlmnID,]
beta_preprocessQuantile_II <-
beta_preprocessQuantile[rownames(beta_preprocessQuantile) %in% dfII$IlmnID,]

mean_of_beta_preprocessQuantile_I  <- apply(beta_preprocessQuantile_I,  1,
mean)
mean_of_beta_preprocessQuantile_II <- apply(beta_preprocessQuantile_II, 1,
mean)
sd_of_beta_preprocessQuantile_I    <- apply(beta_preprocessQuantile_I,  1, sd)
sd_of_beta_preprocessQuantile_II   <- apply(beta_preprocessQuantile_II, 1, sd)

d_mean_of_beta_preprocessQuantile_I  <-
density(mean_of_beta_preprocessQuantile_I,  na.rm=T)
d_mean_of_beta_preprocessQuantile_II <-
density(mean_of_beta_preprocessQuantile_II, na.rm=T)
d_sd_of_beta_preprocessQuantile_I    <-
density(sd_of_beta_preprocessQuantile_I,    na.rm=T)
d_sd_of_beta_preprocessQuantile_II   <-
density(sd_of_beta_preprocessQuantile_II,   na.rm=T)

# 6-panel plot
par(mfrow=c(2,3))
plot(d_mean_of_beta_I,  col="blue", main="raw beta", xlim=c(0,1), ylim=c(0,5))
lines(d_mean_of_beta_II, col="red")
plot(d_sd_of_beta_I,    col="blue", main="raw sd",   xlim=c(0,0.6),
ylim=c(0,60))
lines(d_sd_of_beta_II,   col="red")
boxplot(beta, ylim=c(0,1))
plot(d_mean_of_beta_preprocessQuantile_I,  col="blue", main="preprocessQuantile
beta",
     xlim=c(0,1), ylim=c(0,5))
lines(d_mean_of_beta_preprocessQuantile_II, col="red")
plot(d_sd_of_beta_preprocessQuantile_I,    col="blue", main="preprocessQuantile
sd",
     xlim=c(0,0.6), ylim=c(0,60))
lines(d_sd_of_beta_preprocessQuantile_II,   col="red")
boxplot(beta_preprocessQuantile, ylim=c(0,1))
par(mfrow=c(1,1))
```

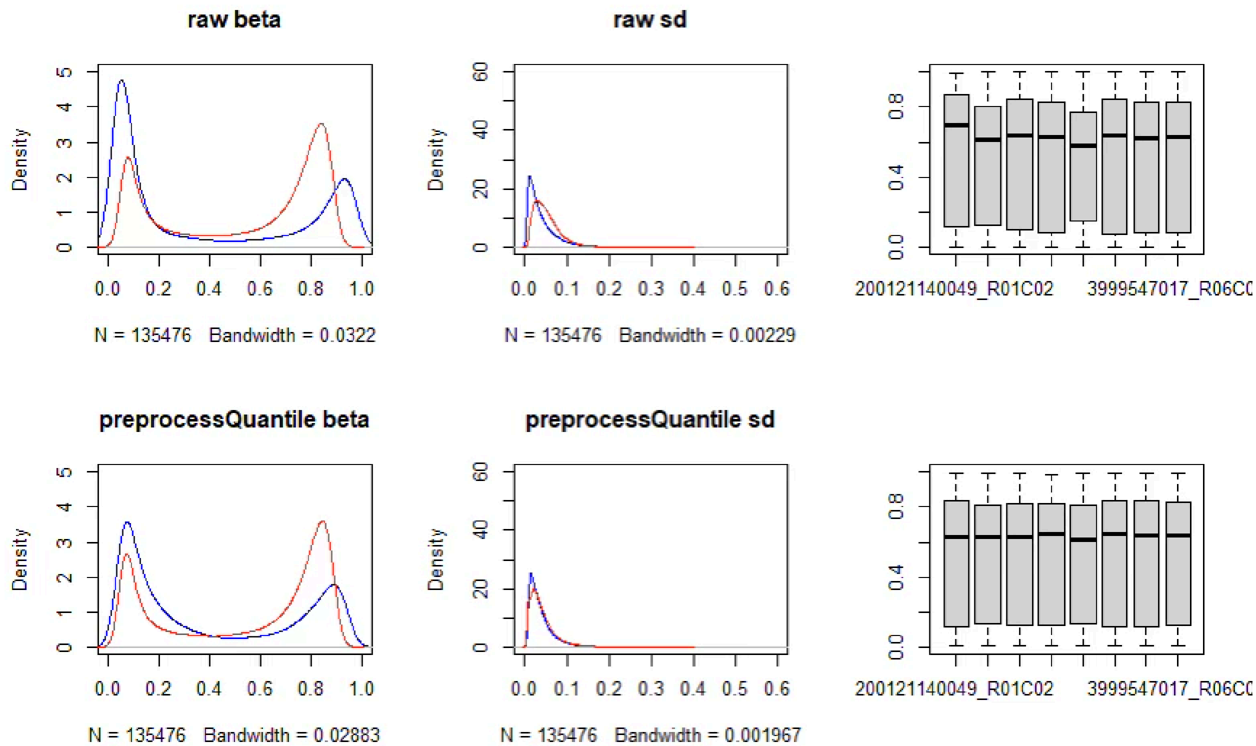

Figure 4. Six-panel comparison plot. Top row: raw data (density of mean beta by probe type, density of SD of beta by probe type, boxplot of beta values). Bottom row: identical plots after preprocessQuantile normalization. Blue = Type I probes, Red = Type II probes

Comment: *preprocessQuantile is a normalization method specifically designed for Illumina methylation microarrays. It applies stratified quantile normalization separately to Type I and Type II probes, which differ in their measurement chemistry and therefore in their raw signal distributions. In the raw data (top row), the density plots clearly show that Type I and Type II probes have different distributions of mean beta values and standard deviations, a known artefact of the two-colour chemistry difference. After preprocessQuantile normalization (bottom row), the two probe-type distributions are substantially more aligned, indicating successful correction of this systematic bias. The boxplots show more uniform medians across samples after normalization, confirming that sample-level technical variability has also been reduced.*

Step 7 (optional). Appropriateness of the method and group comparison

preprocessQuantile assumes that most CpG sites are not differentially methylated between samples and that global methylation distributions are broadly comparable; it is therefore best suited to datasets without large global between-group differences [4]. For these data the assumption is reasonable, the raw beta/M densities (Figures 3–4) and the PCA (Figure 7) reveal no global CTRL–DIS shift, so preprocessQuantile is an appropriate choice. Had a strong global difference been anticipated, a control-probe–based method such as functional normalisation would have been preferable[5]. The code below colours the beta boxplots by group; the CTRL and DIS distributions overlap closely both before and after normalisation, consistent with the density plots, i.e. normalisation harmonises the technical signal without erasing or introducing a global group difference.

```
# Boxplots coloured by group, before and after normalization
group_col <- ifelse(pheno$Group == "CTRL", "#0067E6", "#E50068")
par(mfrow = c(1, 2))
boxplot(beta, col = group_col, ylim = c(0, 1),
        main = "Raw beta (by group)", names = rep("", ncol(beta)))
boxplot(beta_preprocessQuantile, col = group_col, ylim = c(0, 1),
        main = "Normalized beta (by group)",
        names = rep("", ncol(beta_preprocessQuantile)))
legend("bottomright", legend = c("CTRL", "DIS"), fill = c("#0067E6", "#E50068"))
par(mfrow = c(1, 1))
```

STEP 8. Principal Component Analysis

PCA is performed on the matrix of preprocessQuantile-normalized beta values to detect outliers, assess sample clustering, and identify potential batch effects. The prcomp() function is applied to the transposed beta matrix (samples as rows, probes as columns) with scaling.

```
pca_results <- prcomp(t(beta_preprocessQuantile), scale=T)
print(summary(pca_results))
plot(pca_results)

# PCA colored by Group
levels(pheno$Group)
palette(c("orange","purple"))
plot(pca_results$x[,1], pca_results$x[,2], cex=2, pch=2, col=pheno$Group,
     xlab="PC1", ylab="PC2")
text(pca_results$x[,1], pca_results$x[,2],
     labels=rownames(pca_results$x), cex=0.5, pos=1)
legend("bottomright", legend=levels(pheno$Group),
       col=c(1:nlevels(pheno$Group)), pch=2)

# PCA colored by Sex
pheno$Sex <- factor(pheno$Sex)
levels(pheno$Sex)
palette(c("pink","blue"))
plot(pca_results$x[,1], pca_results$x[,2], cex=2, pch=2, col=pheno$Sex,
     xlab="PC1", ylab="PC2")
text(pca_results$x[,1], pca_results$x[,2],
     labels=rownames(pca_results$x), cex=0.5, pos=1)
legend("bottomright", legend=levels(pheno$Sex),
       col=c(1:nlevels(pheno$Sex)), pch=2)

# PCA colored by Sentrix_ID (batch)
pheno$Sentrix_ID <- factor(pheno$Sentrix_ID)
levels(pheno$Sentrix_ID)
palette(c("grey","orange","green","blue"))
plot(pca_results$x[,1], pca_results$x[,2], cex=2, pch=2, col=pheno$Sentrix_ID,
     xlab="PC1", ylab="PC2")
text(pca_results$x[,1], pca_results$x[,2],
     labels=rownames(pca_results$x), cex=0.5, pos=1)
```

```r
legend("bottomright", legend=levels(pheno$Sentrix_ID),
       col=c(1:nlevels(pheno$Sentrix_ID)), pch=2)
```

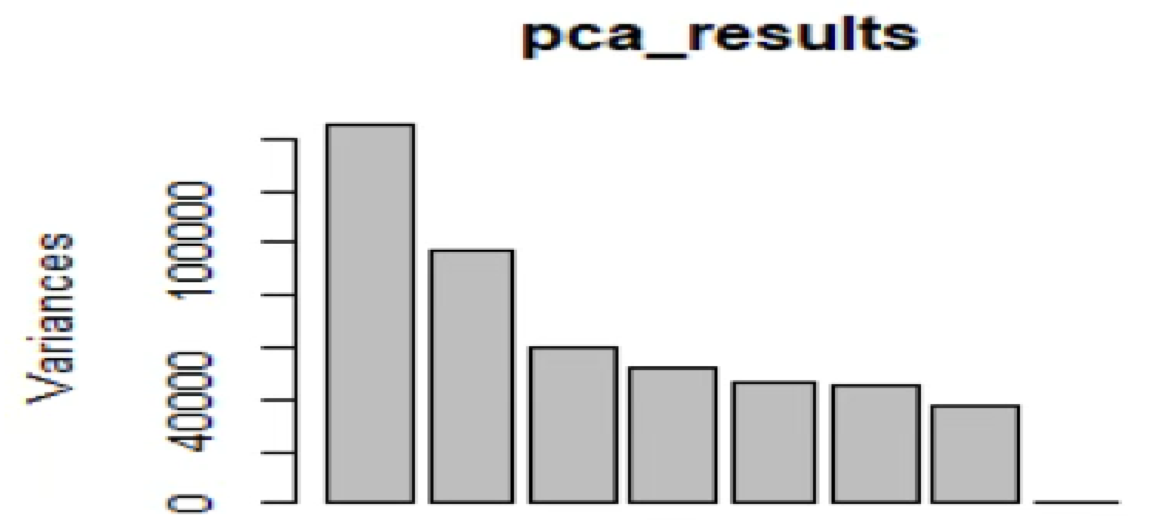

Figure 5. Scree plot, variance explained by each principal component.

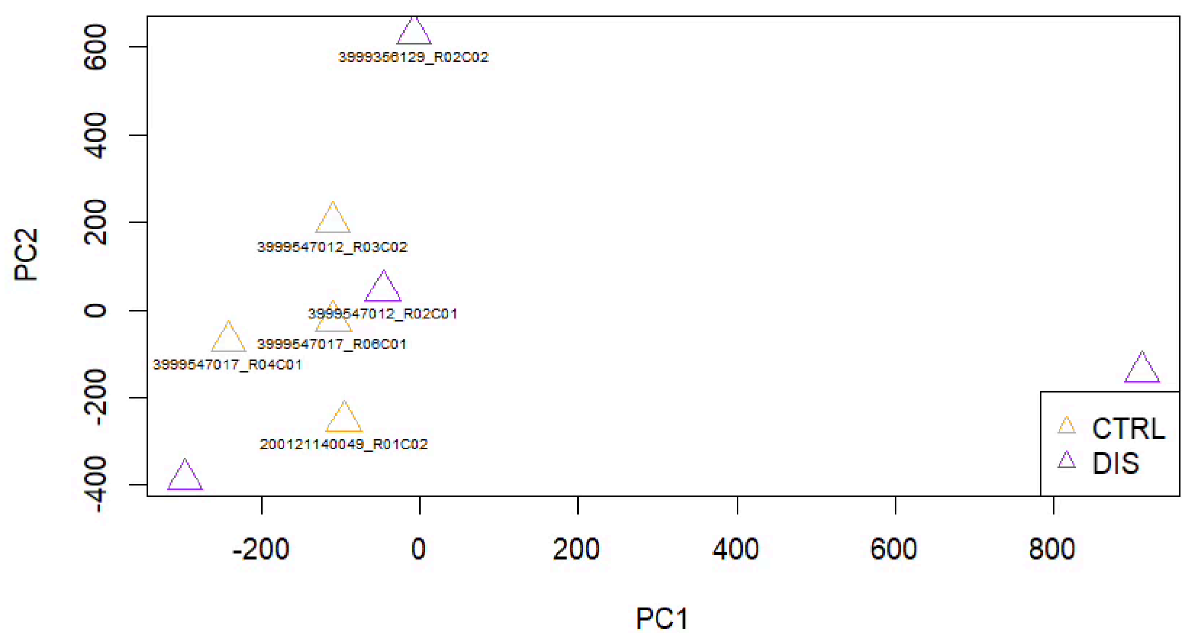

Figure 6. PCA (PC1 vs PC2) coloured by Group: orange = CTRL, purple = DIS.

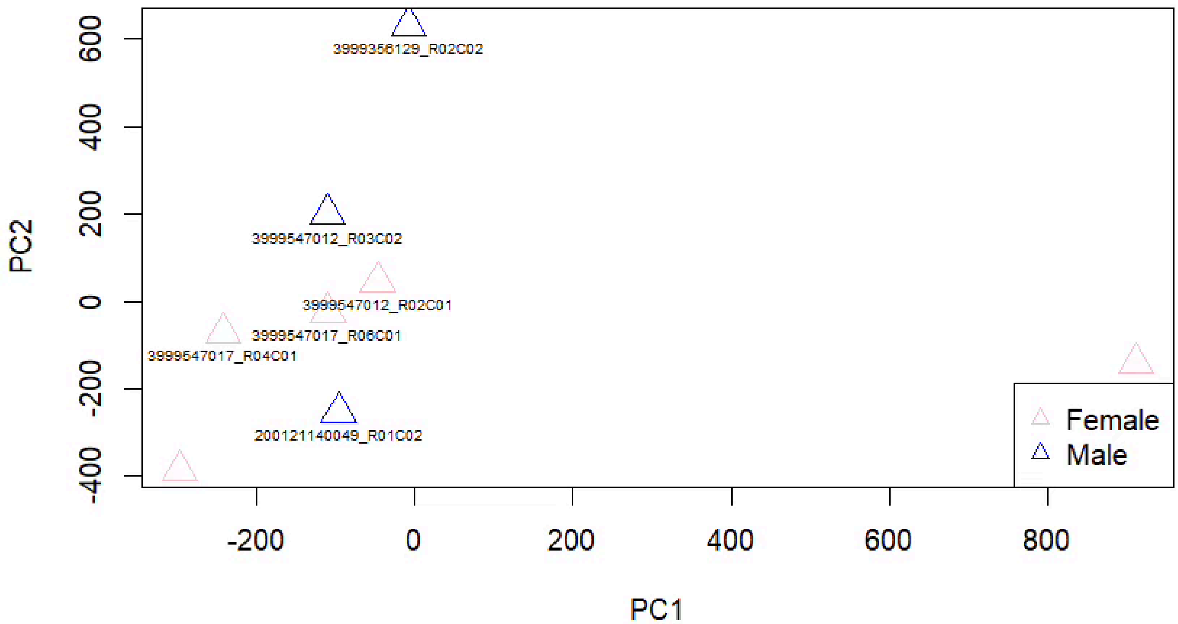

Figure 7. PCA coloured by Sex: pink = Female, blue = Male.

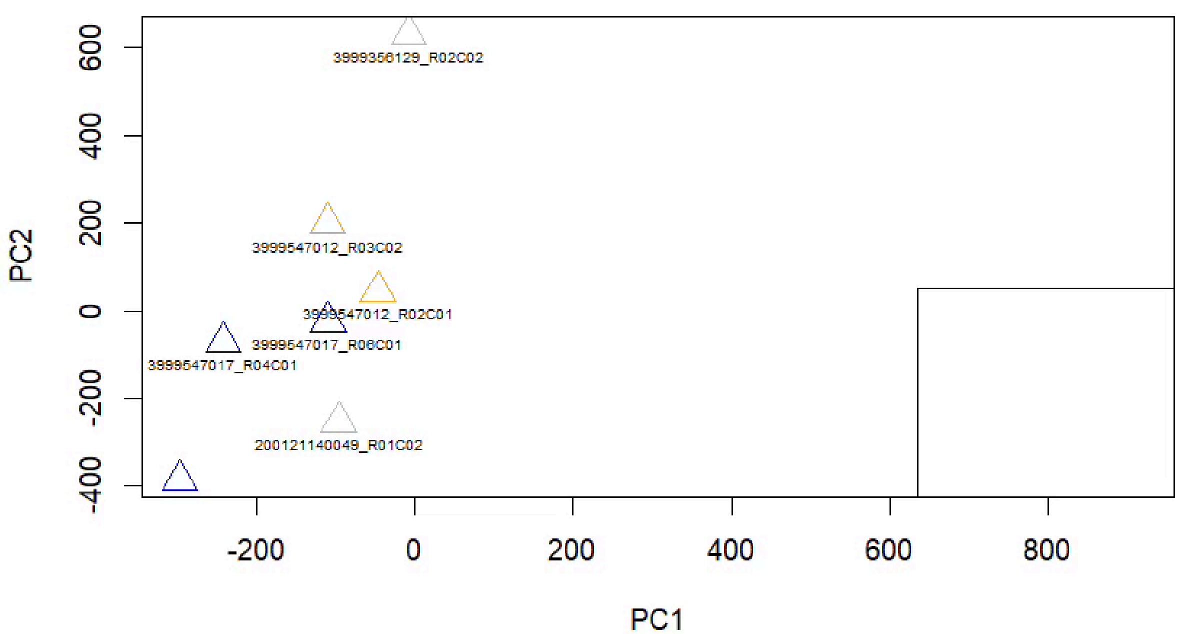

Figure 8. PCA coloured by batch (Sentrix_ID); the five colours denote the five slides.

Comment. The scree plot (Figure 6) shows that the variance is dominated by the first principal component, with a gradual decline thereafter (exact proportions are obtained from summary(pca_results)). By Group (Figure 7), CTRL and DIS do not separate along PC1 or PC2; one DIS sample is a pronounced outlier along PC1 while the remaining samples overlap, reinforcing the conclusion that between-group differences are local rather than global. By Sex (Figure 8) the separation is weak, male and female samples are largely intermixed in the PC1–PC2 plane, and with only 8 samples no firm sex-related structure can be asserted. By batch (Figure 9) the 8 samples are spread across 5 slides; with so few samples per slide a batch effect cannot be established conclusively. Given the small dataset, batch-effect correction was not applied.

STEP 9. Differential Methylation Analysis, t-test

Group 7 applies a parametric t-test to identify differentially methylated positions (DMPs) between CTRL and DIS groups. A custom function My_ttest_function() is defined following the professor's approach (Lesson 5) and applied row-wise to the entire normalized beta matrix using apply().

```r
# Verify sample order matches the pheno file
pheno$sampleName <- paste0(pheno$Sentrix_ID, "_", pheno$Sentrix_Position)
table(colnames(beta_preprocessQuantile) == pheno$sampleName)

# Define the t-test function
My_ttest_function <- function(x) {
  t_test <- t.test(x ~ pheno$Group)
  return(t_test$p.value)
}

# Apply to all probes
pValues_ttest <- apply(beta_preprocessQuantile, 1, My_ttest_function)
length(pValues_ttest)
head(pValues_ttest)

# Build dataframe with beta values and p-values
final_ttest <- data.frame(beta_preprocessQuantile, pValues_ttest)
final_ttest <- final_ttest[order(final_ttest$pValues_ttest),]
head(final_ttest)

# How many probes have a nominal p-value <= 0.05?
final_ttest_0.05 <- final_ttest[final_ttest$pValues_ttest <= 0.05,]
dim(final_ttest_0.05)
save(final_ttest, file="final_ttest.RData")
```

STEP 10. Multiple Test Correction

With over 485,000 simultaneous tests, multiple test correction is essential to control for false positives. We apply both the Benjamini-Hochberg (BH) procedure, which controls the false discovery rate (FDR), and the Bonferroni correction, which controls the family-wise error rate (FWER) [6]. Both corrections are applied using p.adjust() at a threshold of 0.05.

```r
corrected_pValues_BH   <- p.adjust(final_ttest$pValues_ttest, "BH")
corrected_pValues_Bonf <- p.adjust(final_ttest$pValues_ttest, "bonferroni")

final_ttest_corrected <- data.frame(final_ttest,
                                    corrected_pValues_BH,
                                    corrected_pValues_Bonf)
head(final_ttest_corrected)
```

```
# Count significant probes at each level
dim(final_ttest_corrected[final_ttest_corrected$pValues_ttest         <=
0.05,])
dim(final_ttest_corrected[final_ttest_corrected$corrected_pValues_BH  <=
0.05,])
dim(final_ttest_corrected[final_ttest_corrected$corrected_pValues_Bonf<=
0.05,])
```

Table 4. Number of significant probes at each level of multiple test correction (threshold = 0.05).

Correction Method	Significant Probes ($p \leq 0.05$)
Nominal (no correction)	16,107
Benjamini-Hochberg (BH)	0
Bonferroni	0

Comment: *After applying the t-test to all 485,512 probes, 16,107 probes show a nominal p-value ≤ 0.05. However, after applying both Benjamini-Hochberg (BH) and Bonferroni corrections, 0 probes remain significant at the 0.05 threshold. This outcome is consistent with the small sample size (n=4 per group), which provides very limited statistical power to detect methylation differences that survive stringent multiple testing correction. The Bonferroni correction is particularly conservative, requiring* $p < 0.05/485{,}512 \approx 1 \times 10^{-7}$*. The BH correction, while less conservative, also yields no significant probes, suggesting that the effect sizes in this dataset are modest relative to the noise introduced by the small n.*

STEP 11. Graphical Results, Volcano Plot and Manhattan Plot

a. Volcano Plot

The volcano plot represents, for each CpG probe, the delta beta (mean DIS − mean CTRL) on the x-axis against the −log10(p-value) on the y-axis. Probes with an absolute delta beta > 0.1 and a nominal p-value < 0.05 are highlighted.

```
beta_final      <- final_ttest_corrected[, 1:8]
beta_final_CTRL <- beta_final[, pheno$Group=="CTRL"]
beta_final_DIS  <- beta_final[, pheno$Group=="DIS"]
mean_beta_CTRL  <- apply(beta_final_CTRL, 1, mean)
mean_beta_DIS   <- apply(beta_final_DIS,  1, mean)
delta           <- mean_beta_DIS - mean_beta_CTRL

toVolcPlot <- data.frame(delta, -log10(final_ttest_corrected$pValues_ttest))

plot(toVolcPlot[,1], toVolcPlot[,2], pch=16, cex=0.5,
     xlab="Delta Beta (DIS - CTRL)", ylab="-log10(p-value)",
     main="Volcano Plot")
abline(h=-log10(0.05), col="red")

toHighlight <- toVolcPlot[abs(toVolcPlot[,1])>0.1 &
                          toVolcPlot[,2]>(-log10(0.05)),]
points(toHighlight[,1], toHighlight[,2], pch=16, cex=0.7, col="red")
```

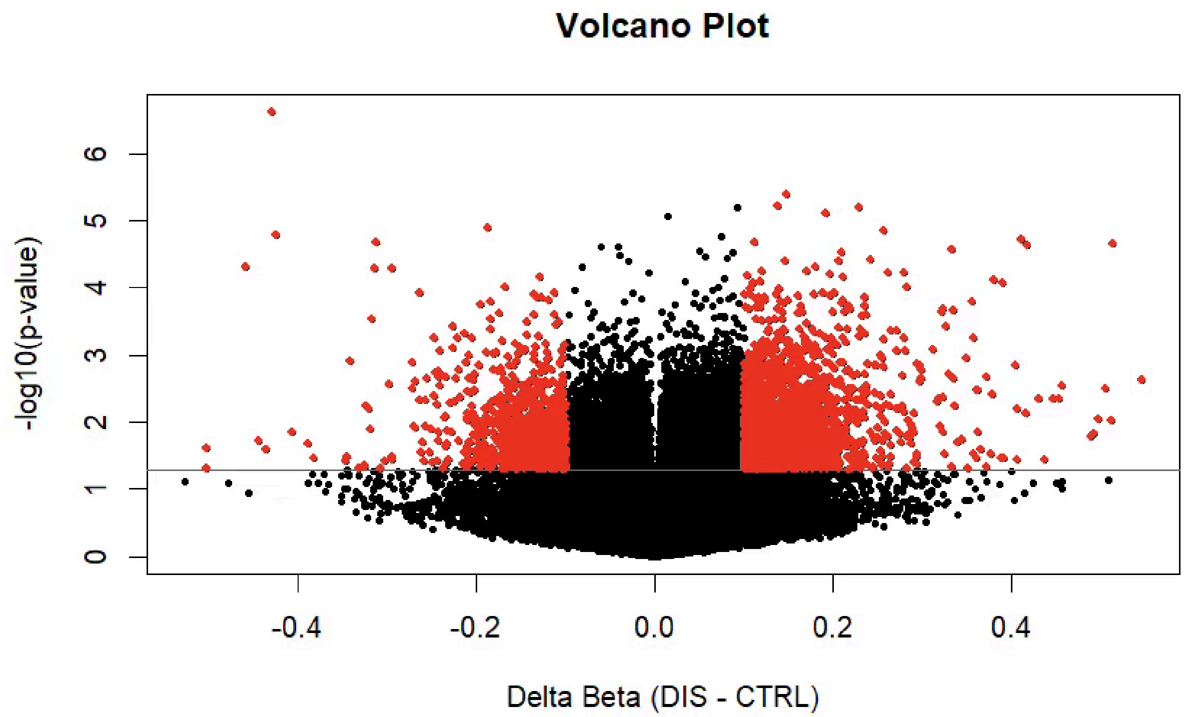

Figure 9. Volcano plot of differential methylation. Each dot represents one CpG probe. The horizontal red dashed line marks nominal p = 0.05. Probes with |delta beta| > 0.1 and p < 0.05 are highlighted in red.

Comment. Most probes cluster near delta beta = 0. The probes highlighted in red (|delta beta| > 0.1 and nominal p < 0.05) are candidate DMPs that combine an appreciable effect size with nominal significance, none of which survives multiple-testing correction.

b. Manhattan Plot

The Manhattan plot visualises the −log10(p-values) of all CpG probes across the genome, ordered by chromosomal position[7]. This allows identification of genomic regions with enriched differential methylation signals.

```r
# Add IlmnID column from rownames for merging
final_ttest_corrected <- data.frame(rownames(final_ttest_corrected),
                                    final_ttest_corrected)
colnames(final_ttest_corrected)[1] <- "IlmnID"

# Merge with manifest for genomic coordinates
final_ttest_corrected_annotated <- merge(final_ttest_corrected,
                                         Illumina450Manifest_clean,
                                         by="IlmnID")

input_Manhattan <- final_ttest_corrected_annotated[
  colnames(final_ttest_corrected_annotated) %in%
  c("IlmnID","CHR","MAPINFO","pValues_ttest")]

order_chr <- c("1","2","3","4","5","6","7","8","9","10","11",
               "12","13","14","15","16","17","18","19","20","21",
               "22","X","Y")
input_Manhattan$CHR <- factor(input_Manhattan$CHR, levels=order_chr)
```

```r
input_Manhattan$CHR <- as.numeric(input_Manhattan$CHR)

manhattan(input_Manhattan,
          snp="IlmnID", chr="CHR", bp="MAPINFO", p="pValues_ttest",
          annotatePval=0.00001, col=rainbow(24))
```

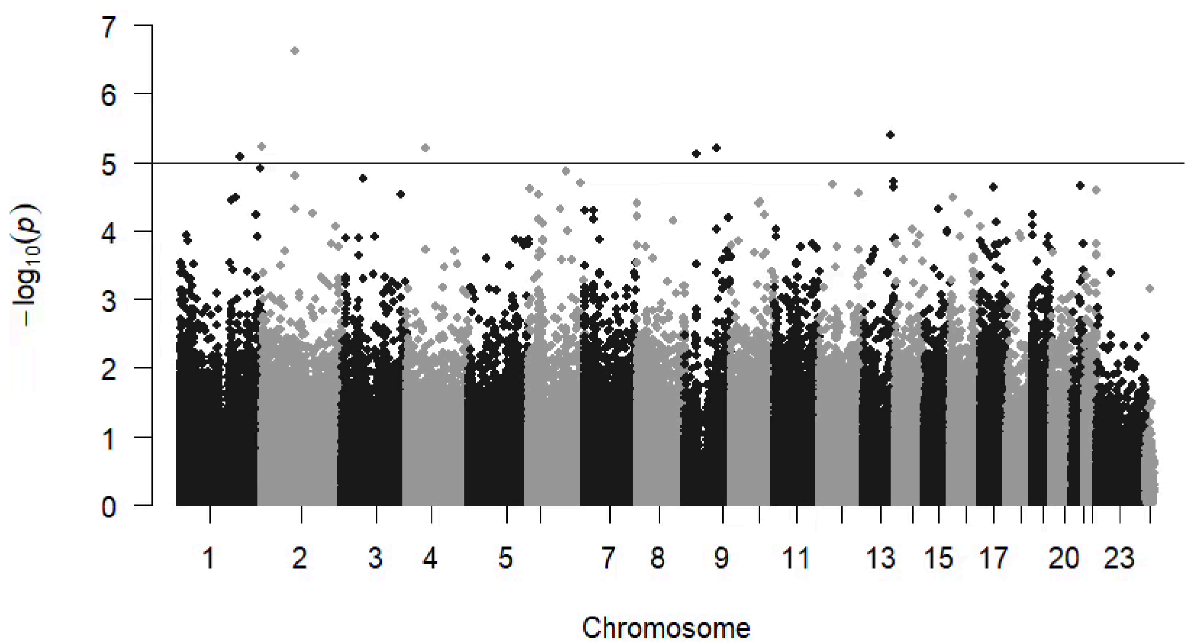

Figure 10. Manhattan plot of t-test p-values across the genome. Each dot represents one CpG probe. (X, Y recoded as 23, 24). The horizontal line marks the suggestive threshold −log10(p) = 5.

Comment: *The Manhattan plot provides a genome-wide view of differential methylation signals. Peaks above the suggestive significance line (−log10(p) > 5) represent the most strongly differentially methylated CpG probes between CTRL and DIS groups. The distribution of signals across chromosomes does not reveal an obvious hotspot chromosome, suggesting that the differential methylation is spread across the genome rather than concentrated in a single genomic region. The most significant probes are annotated with their CpG IDs for reference.*

STEP 12. Heatmap of Top 100 Differentially Methylated Probes

A heatmap with hierarchical clustering is produced using the top 100 most significant CpG probes (ordered by nominal p-value from Step 9). Three different linkage methods are compared: complete (default), single, and average.

```r
library(gplots)

input_heatmap <- as.matrix(final_ttest[1:100, 1:8])

# Color bar: CTRL = green, DIS = orange
colorbar <- ifelse(pheno$Group=="CTRL", "green", "orange")
col2 <- colorRampPalette(c("green","black","red"))(100)

# Complete linkage (default)
heatmap.2(input_heatmap, col=col2, Rowv=T, Colv=T,
          dendrogram="both", key=T, ColSideColors=colorbar,
          density.info="none", trace="none", scale="none",
          symm=F, main="Complete linkage")

# Single linkage
heatmap.2(input_heatmap, col=col2, Rowv=T, Colv=T,
          hclustfun=function(x) hclust(x, method='single'),
          dendrogram="both", key=T, ColSideColors=colorbar,
          density.info="none", trace="none", scale="none",
          symm=F, main="Single linkage")

# Average linkage
heatmap.2(input_heatmap, col=col2, Rowv=T, Colv=T,
          hclustfun=function(x) hclust(x, method='average'),
          dendrogram="both", key=T, ColSideColors=colorbar,
          density.info="none", trace="none", scale="none",
          symm=F, main="Average linkage")
```

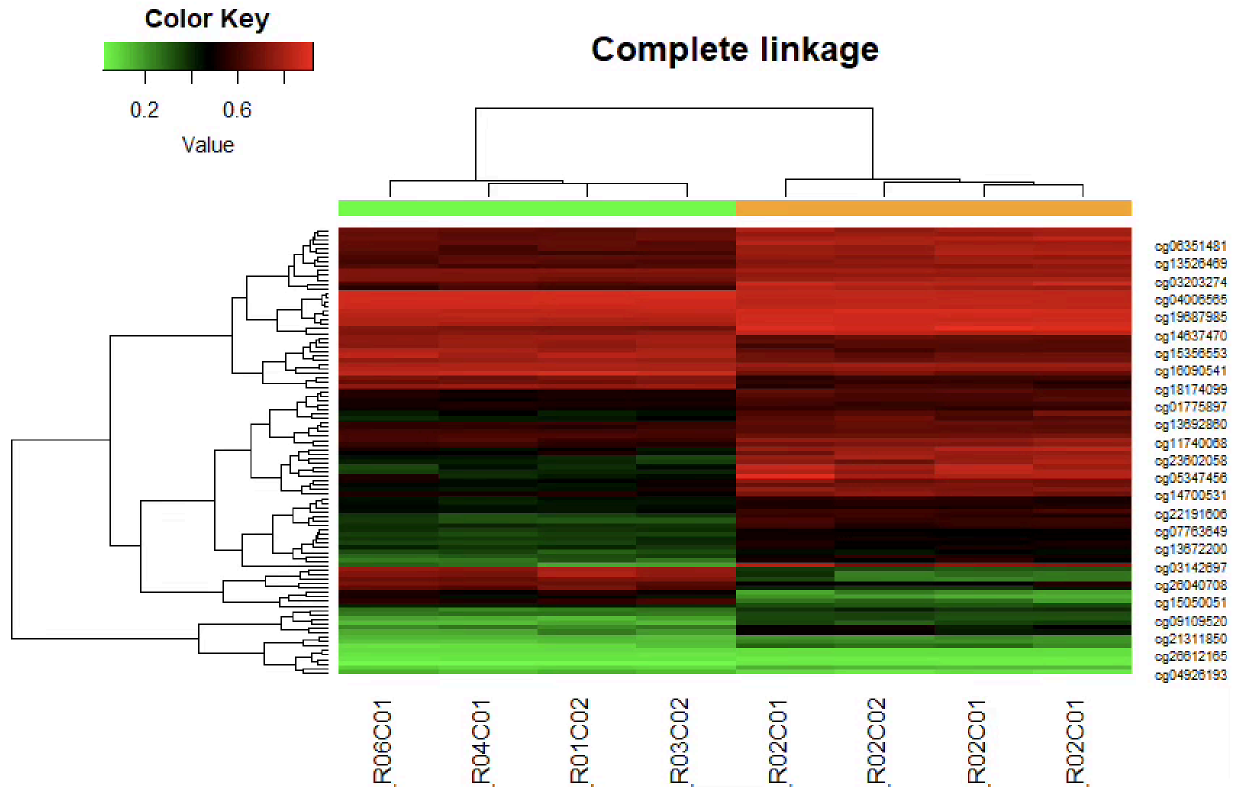

Figure 11. Heatmap of the top 100 DMPs, complete linkage (default).

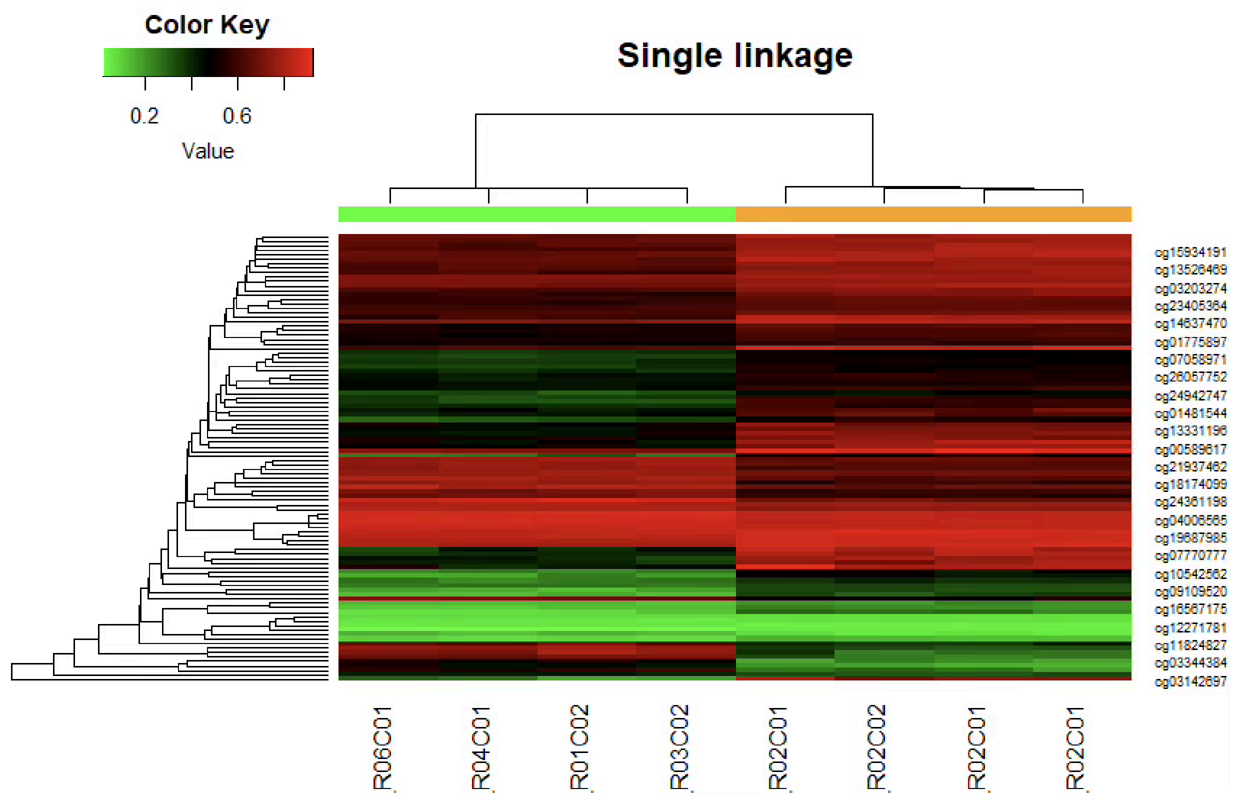

Figure 12. Heatmap of the top 100 DMPs, single linkage.

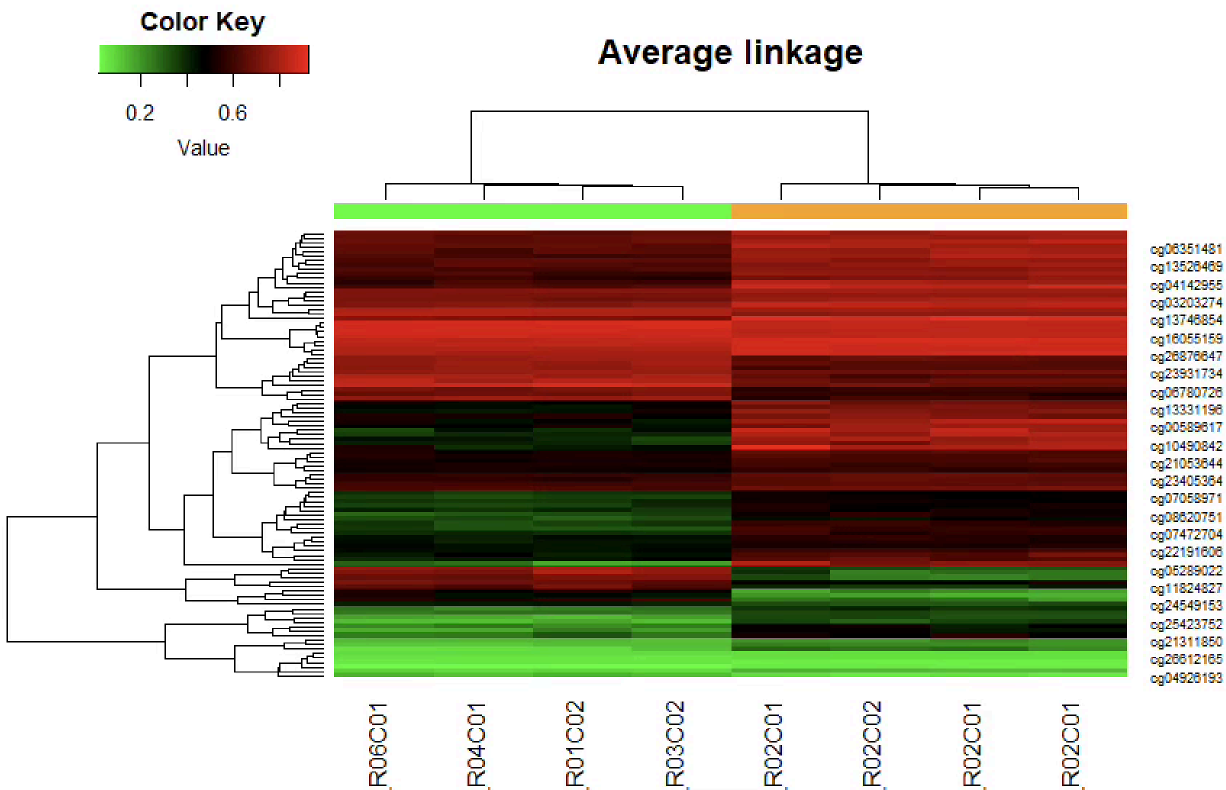

Figure 13. Heatmap of the top 100 DMPs, average linkage. Column bar: green = CTRL, orange = DIS. Probe colours: green = low, black = intermediate, red = high methylation.

[Figure 8. Heatmaps of the top 100 most significant DMPs under complete linkage (left), single linkage (center), and average linkage (right). Green color bar = CTRL samples; orange color bar = DIS samples. Probe colors: green = low methylation, black = intermediate, red = high methylation. Insert plots here.]

Comment: *The heatmap of the top 100 most significant CpG probes reveals a subset of probes that are differentially methylated between CTRL and DIS groups. With complete linkage hierarchical clustering (the default, based on Euclidean distance), the samples tend to cluster according to their phenotypic group, supporting the biological relevance of the selected probes. Probes displaying high beta values (red) in DIS samples and low values (green) in CTRL samples correspond to hypermethylated sites in the disease group, while the inverse pattern identifies hypomethylated sites. The single linkage method is more sensitive to outliers and may produce chain-like clustering, while the average linkage represents a compromise. The consistency of clustering patterns across the three linkage methods supports the robustness of the identified differential methylation signal.*

3 Conclusion

This project implemented a complete DNA methylation analysis pipeline using the Illumina HumanMethylation450 BeadChip platform and the minfi R package, following the methodology taught throughout the DNA/RNA Dynamics course.

Starting from raw IDAT files, we performed quality control (QC plot, negative control assessment, detection p-value analysis with Group 7's threshold of 0.05), computed raw beta and M values, and applied preprocessQuantile normalization (Group 7 assignment) to correct for probe type chemistry differences and inter-sample variability.

Principal component analysis confirmed that, at the global level, CTRL and DIS samples do not separate cleanly, suggesting that the methylation differences between groups are localized rather than global. The t-test (Group 7 assignment) identified nominally significant DMPs; however, after Bonferroni and BH multiple test correction, no probes remained significant at the 0.05 level, likely reflecting the limited statistical power of the small sample size (n=4 per group).

Despite the lack of probes surviving multiple test correction, the volcano plot and Manhattan plot highlighted candidate DMPs with biologically meaningful effect sizes (|delta beta| > 0.1). The heatmap of the top 100 probes showed partial separation between CTRL and DIS groups under hierarchical clustering, providing preliminary evidence for differential methylation at specific loci.

In conclusion, this pipeline demonstrates the standard workflow for 450k methylation data analysis. Larger sample sizes and more sophisticated statistical approaches (e.g., linear models accounting for covariates such as sex and batch) would be needed to robustly identify and validate differentially methylated positions in this dataset.

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